CSE-476: Introduction to Natural Language Processing
Quiz 2

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. Which option describes a model that learns the data distribution for classification?

A. Generative models model $P(x, y)$, while discriminative models model $P(y \mid x)$
B. Discriminative models learn the data distribution and use that knowledge for classification
C. Discriminative models require no training data
D. Generative models are less explainable

Q2. Which technique converts raw term counts into importance scores for words in documents?

A. Laplace smoothing
B. Regularization
C. TF-IDF
D. Softmax normalization

Q3. Which option describes a method that fails when classes are not linearly separable?

A. It cannot use vector representations
B. It only works for linearly separable data
C. It requires probabilistic outputs
D. It cannot be trained with gradient descent

Q4. In language models, which option prevents unseen words from having probability zero?

A. It improves cosine similarity
B. It reduces dimensionality
C. It avoids zero probabilities for unseen words
D. It speeds up gradient descent

Q5. Which option describes an algorithm that updates parameters using the full dataset each step?

A. SGD updates weights using the entire dataset every step
B. SGD updates weights using one or a subset of training examples
C. GD converges faster because updates are noisier
D. SGD cannot be used with neural networks

Q6. Which option states that similar meanings arise from words sharing similar contexts?

A. Words are represented by their dictionary definitions
B. Words that appear in similar contexts have similar meanings
C. Words are clustered by part of speech
D. Words are ranked by frequency

Q7. Which option selects the class with the largest raw score without any normalization?

A. To penalize large model weights
B. To reduce feature dimensionality
C. To select the class with the largest raw score without normalization
D. To convert raw class scores into a valid probability distribution

Q8. In online learning, which option updates model parameters after each training example?

A. After every training example
B. Only when the prediction is correct
C. Only when the prediction is incorrect
D. Only after one full pass over the data

Q9. Which option best describes a scheme that emphasizes words frequent in a document but rare overall?

A. Increase the weight of very common words
B. Represent word order
C. Convert sparse vectors into dense embeddings
D. Highlight words that are frequent in a document but rare across documents

Q10. Which option describes a measure that boosts the influence of very frequent words?

A. It ignores vector length while calculating similarity
B. It increases the influence of high-frequency words
C. It converts similarity scores into probabilities
D. It reduces the dimensionality of word embeddings